

Distributed Adiabatic Ground State Estimation of 3D Lattices via a Patch-and-Kick Volumetric Engine

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July 4, 2026

Abstract

The simulation of large-scale, strongly correlated quantum systems is profoundly limited by the exponential scaling of the Hilbert space. In this work, we present a high-throughput volumetric engine for the distributed simulation of an $SU(2)$ 3D Transverse-Field Ising Model (TFIM) undergoing adiabatic annealing. To overcome the memory and computational constraints of simulating global entanglement, we employ a “patch-and-kick” mechanism: the lattice is partitioned into isolated sub-volumes (patches) simulated natively on multi-GPU architectures, with entanglement across boundaries approximated by iterative, classical macro-spin exchanges (kicks) acting on the boundary qubits. This approach draws inspiration from mean-field decouplings, holographic traversable wormhole protocols, and Sachdev-Ye-Kitaev (SYK) models, providing a classical proxy for quantum correlations. We detail the theoretical framework, the Strang-splitting numerical implementation, and evaluate the algorithm’s performance using Cross-Entropy Benchmarking (XEB) and Heavy Output Generation (HOG) metrics.

1 Introduction

The quest to find the ground state of generic local Hamiltonians is a central problem in condensed matter physics, quantum chemistry, and computational complexity, generally classified as QMA-hard. Quantum Adiabatic Evolution (QAE) and quantum annealing [?] provide a physical framework to reach these ground states by slowly evolving the system from a trivial, easily prepared ground state to the target Hamiltonian’s ground state. However, on near-term hardware or full-state classical simulators, system size is strictly bottlenecked by qubit counts and exponentially growing memory requirements.

In recent years, alternative quantum-inspired classical simulations have emerged, drawing connections between condensed matter models and holographic dualities. Notably, models such as the Sachdev-Ye-Kitaev (SYK) model have been utilized to study maximal scrambling and traversable wormhole dynamics [?, ?]. In these holographic setups, distinct conformal boundaries are coupled via weak, classical multi-qubit interactions, which functionally operate as teleportation or information exchange channels across boundaries.

Inspired by these protocols, we introduce the ThereminQ High-Performance Computing (HPC) Volumetric Engine. We circumvent the exponential memory barrier by physically partitioning a 3D lattice into smaller $SU(2)$ sub-volumes (patches) distributed across an array of GPUs. Entanglement generation within a patch is simulated exactly via PyQrack [?], while inter-patch entanglement is approximated dynamically. We introduce a “patch-and-kick” mechanism, where boundary qubits are subjected to classical, mean-field-like rotational kicks derived from the average Bloch vectors of adjacent patches. We analyze the effectiveness of this proxy using metrics native to quantum volume verification, primarily Cross-Entropy Benchmarking (XEB) [?] and Heavy Output Generation (HOG).

2 Theoretical Framework

We consider the 3D Transverse-Field Ising Model (TFIM) coupled to a longitudinal field, whose Hamiltonian is given by:

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h_x \sum_i X_i - h_z \sum_i Z_i \quad (1)$$

where X_i, Z_i are Pauli matrices, J is the exchange coupling, and h_x, h_z are the transverse and longitudinal magnetic fields respectively.

To find the ground state, we employ an adiabatic schedule parameterized by $s(t) = t/T \in [0, 1]$, interpolating between a non-interacting initial Hamiltonian H_0 and the target Hamiltonian H_{target} :

$$H(s) = (1 - s)H_0 + sH_{\text{target}} \quad (2)$$

In our implementation, the initial state is driven by a strong transverse field $h_x(0) \gg 1$, such that the initial state is approximately $|+\rangle^{\otimes N}$.

For a system partitioned into discrete patches $\mathcal{P}_k$, the total Hamiltonian is strictly non-separable. We approximate the inter-patch boundary Hamiltonian $H_{\text{boundary}} = -g_{\text{face}} \sum_{\langle i \in \partial\mathcal{P}_a, j \in \partial\mathcal{P}_b \rangle} \sigma_i \cdot \sigma_j$ using a mean-field surrogate. For a boundary qubit $i \in \partial\mathcal{P}_a$, the interaction with the adjacent face $\partial\mathcal{P}_b$ is replaced by a coupling to the macroscopic magnetization of $\partial\mathcal{P}_b$:

$$\langle \sigma_{\partial\mathcal{P}_b} \rangle = \frac{1}{|\partial\mathcal{P}_b|} \sum_{j \in \partial\mathcal{P}_b} \langle \sigma_j \rangle \quad (3)$$

The boundary interaction energy is thus minimized via iterative classical kicks on the boundary qubits, scaling with a coupling constant g_{face} , acting as a classical proxy for quantum entanglement.

3 Numerical Implementation

The distributed simulation is orchestrated across multiple GPU workers. Each worker maintains a persistent PyQuack simulator state for a sub-volume of size $L_x \times L_y \times L_z$ (e.g., $3 \times 3 \times 2 = 18$ qubits).

3.1 Strang Splitting Evolution

Intra-patch time evolution is performed using a symmetric Trotter-Suzuki decomposition, specifically a second-order Strang splitting [?]. For a time step Δt , the unitary evolution operator $U(\Delta t) = e^{-iH(s)\Delta t}$ is approximated as:

$$U(\Delta t) \approx e^{-i\frac{\Delta t}{2}h_x \sum X_i} e^{-i\frac{\Delta t}{2}h_z \sum Z_i} e^{-i\Delta t J \sum Z_i Z_j} e^{-i\frac{\Delta t}{2}h_z \sum Z_i} e^{-i\frac{\Delta t}{2}h_x \sum X_i} \quad (4)$$

The $Z_i Z_j$ interaction is executed via CNOT gates conjugating an $R_z(-2J\Delta t)$ rotation.

3.2 The Patch-and-Kick Mechanism

Following the intra-patch unitary evolution, the inter-patch coupling is simulated. The process iterates as follows:

1. **Bloch Vector Extraction:** Without projecting the state (via clone-free internal state querying), the local expectation values $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$ are measured for all boundary qubits in every patch.

2. **Macroscopic Kick Calculation:** For adjacent patches, the mean Bloch vector of the opposing boundary face is computed.
3. **Kick Application:** The accumulated classical correlations are applied as local $SU(2)$ rotations $R_x(k_x), R_y(k_y), R_z(k_z)$ on the boundary qubits, where $k_\alpha \propto g_{\text{face}} \langle \sigma_\alpha \rangle$.

4 Results & Benchmarking

To evaluate the fidelity and scrambling characteristics of the simulation, we compute Cross-Entropy Benchmarking (XEB) and Heavy Output Generation (HOG) at the conclusion of the anneal. XEB has become the gold standard for verifying quantum computational advantage and deep circuit entanglement, notably utilized by the Google Quantum AI team [?].

The linear XEB fidelity is estimated by comparing the sampled bitstrings against the ideal un-kicked bulk probabilities:

$$\mathcal{F}_{\text{XEB}} = \frac{\sum_i (p_{\text{ideal}}(x_i) - 1/D)(p_{\text{meas}}(x_i) - 1/D)}{\sum_i (p_{\text{ideal}}(x_i) - 1/D)^2} \quad (5)$$

where $D = 2^N$ is the Hilbert space dimension. High XEB values signify that the local evolution correctly populates the heavy probability states, while the boundary purity metric ($P = \langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2$) monitors the pseudo-decoherence induced by the non-unitary classical boundary kicks.

Preliminary tracking of the macroscopic boundary energy confirms that the patch-and-kick mechanism effectively lowers the total energy profile, steering the composite system closer to the global true ground state without necessitating an exponentially large unified state vector.

5 Discussion

We have demonstrated a highly scalable, distributed volumetric engine capable of estimating 3D lattice ground states. The patch-and-kick protocol offers a compelling compromise between local exact unitary evolution and global classical mean-field approximations. This approach bridges concepts from quantum-inspired machine learning and holographic SYK traversable wormholes, mapping the abstract concept of boundary interaction to functional, classical multi-GPU messaging. Future work will investigate the transition limits of g_{face} and how dynamically updated neural-network ansätze could replace the rigid mean-field boundary kicks.

References

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